Mathematics - K-12 Level Assessment 4 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. The value of the digit 5 in 24,513 is how many times the value of the digit 5 in 357?
   (a) 10
   (b) 100
   (c) 1,000
   (d) 10,000
2. Which expression is equivalent to 5 x 9?
   (a) (5 x 4) x (6 x 5)
   (b) (5 x 5) + (5 x 4)
   (c) (5 x 5) + (5 x 9)
   (d) (5 x 9) x (6 x 9)
3. Which among the following would result in the narrowest confidence interval?
   (a) Small sample size and 95% confidence
   (b) Small sample size and 99% confidence
   (c) Large sample size and 95% confidence
   (d) Large sample size and 99% confidence
4. A number's prime factors are 2, 5, 7, 13, and 31. Which of the following must be a factor of the number?
   (a) 4
   (b) 6
   (c) 10
   (d) 15
5. What is the product of the greatest even prime number and the least odd prime number?
   (a) 6
   (b) 9
   (c) 12
   (d) 14

True / False (10 marks)

Answer True or False

6. True or False: The total amount spent on lunch, including a 20% tip on a $44meal, is53.80.
7. True or False: The median of the data set 79.88,21.88, 29.88,19.88, 17.88is19.88.
8. True or False: Micah ate an odd number of apples and an odd number of strawberries yesterday.
9. True or False: There are 31 integers from -200 to 200 that are congruent to 5 modulo 13.
10. True or False: The total number of chairs set up for the meeting is 36, which is the result of adding the rows and chairs together.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. The sequence $\{a_n\}$ satisfies $a_1 = 1$ and $5^{a_{n+1}-a_n} - 1 = \frac{1}{n+\frac{2}{3}}$ for $n \geq 1$. Find the least integer k greater than 1 for which a_k is an integer.
12. Discuss how you would determine the number of basketball teams that can be formed from 30 players, given that each team consists of 5 players. Provide a detailed explanation of your reasoning.

End of Paper